SUJATA NUTRILIVE AGROTECH

OFFICIAL REGULATORY COMPLIANCE & 3-DOMAIN LABORATORY QUALITY DOSSIER

1. STATUTORY ENTITIES, CONTRACT MANUFACTURING & ACCREDITED TESTING

Brand Owner & Marketer (FBO):	Sujata Nutrilive Agrotech (Prop. Yash Rajendra Surywanshi) Plot No 125, Sai Niwas, ITI Ambad Link Road, Trimurti Chowk, Kamatwada, Nashik, Maharashtra - 422008 **FSSAI Registration:** #21526027000179 (Govt of Maharashtra FDA | Category 13: Nutritional Uses)
Contract Manufacturer:	**Spiru Swastha India Pvt Ltd** | Survey No: 125/2, Block No. 118, Moje Vasrav Village, Mangrol, Surat, Gujarat - 394421 **FSSAI License:** #21526027000179 | Specialization: Bio-secure hydro-dome spirulina cultivation & sub-40 deg C cryo-thermal dehydration
Testing Laboratory:	**Lilaba Analytical Laboratories** (Surat, Gujarat) **FSSAI Notified Laboratory Reg:** #52/W/FSSAI/2018 | **NABL Accreditation:** Certificate #TC-7179 (Valid through 19 October 2027)

2. 3-DOMAIN SCIENTIFIC ASSAY ARCHITECTURE & BATCH TRACEABILITY

Assay Domain 01 **Master Formulation Baseline**	**Report #LAL-NN-121023002 (Lilaba NABL Scope TC-7179):** Foundation proximate assay establishing the empirical botanical baseline: 62.3% protein (IS 7219 Kjeldahl), 291.15 mg/kg natural iron (FSSAI Metals 5.0), and 398.00 kcal/100g.
Assay Domain 02 **Preserved Bio-Actives**	**Botanical Biomarkers & Enzymes (Sub-40 deg C Cryo-Preserved):** C-Phycocyanin 21.4% (Bennett & Bogorad UV-Vis @ 620nm), Superoxide Dismutase >12,000 U/kg (Marklund Pyrogallol Autoxidation Kinetic Assay), Total Chlorophyll-a 1,150 mg/100g (AOAC 940.03).
Assay Domain 03 **Commercial Release CoA**	**Batch #SJT-2026-B08 Certificate of Analysis (Mfg: Aug 2026, Exp: Jul 2028):** Official release certificate conforming to Master Specification SPEC-SN-2026-REV03 with full ICP-MS heavy metal and microbiological clearance (see Page 2).

3. VERIFIED ANALYTICAL TEST RESULTS (Lilaba Analytical Laboratories NABL Certificate #TC-7179)

Parameter Tested	Measured Result	Analytical Method Protocol	Regulatory & Quality Benchmark
Total Plant Protein	**62.3% w/w**	IS 7219 Protocol	Gold Standard (Min 55.0% FSSAI standard)
Total Iron (as Fe)	**291.15 mg/kg (29.1 mg/100g)**	FSSAI Manual - Metals Method 5.0	Natural Bioavailable Algal Iron
Natural Calcium (as Ca)	**1293.64 mg/kg**	FSSAI Manual - Metals Method 5.0	Natural Bone & Cellular Nutrient
Natural Potassium (as K)	**14598.36 mg/kg**	FSSAI Manual - Metals Method 5.0	Natural Algal Cellular Electrolyte
Natural Vitamin A	**183.12 mcg / 100g**	IS 5886 Protocol	Natural Bioactive Retinoid Precursor
Natural Vitamin C	**12.18 mg / 100g**	IS 5838 Protocol	Endogenous Cellular Antioxidant
Natural Vitamin E	**5.23 mg / 100g**	IS 7235 Protocol	Cellular Membrane Lipid Protection
Dietary Fiber	**3.72%**	IS 11062 Protocol	Prebiotic Algal Matrix
Total Energy	**398.00 Kcal / 100g**	FAO Chap 3, Method 3.5	Nutritive Functional Cellular Fuel
Cholesterol & Trans-Fat	**BLQ / 0.50%**	LAL/CHEM/SOP/FD/GF/18	Below Limit of Quantification (Heart Safe)
Total Mineral Ash	**6.72%**	Gravimetric Protocol	Natural Endogenous Mineral Matrix
Residual Moisture	**3.28%**	Sub-40 deg C Dehydration	Stable Shelf Life & Microbial Safety

4. STATUTORY DECLARATION & DOSSIER INDEX

Master Formulation Benchmark & Commercial Traceability: Report #LAL-NN-121023002 serves as the official Master Formulation Benchmark Assay establishing the empirical proximate nutritional baseline of Sujata's proprietary Arthrospira platensis cultivation strain and sub-40 deg C gentle dehydration process (yielding 62.3% plant protein under IS 7219 and 291.15 mg/kg natural iron under FSSAI Metals Method 5.0). All commercial production batches are contract-manufactured by Spiru Swastha India Pvt Ltd (Surat, Gujarat) to this identical benchmark specification and marketed by Sujata Nutrilive Agrotech under Government of Maharashtra FSSAI Registration #21526027000179.

Document Index: Page 1: Master Executive Compliance Dossier | Page 2: Commercial Batch CoA (Batch #SJT-2026-B08) | Page 3: Botanical Biomarkers & Scientific Demarcation | Pages 4-5: Lilaba Analytical Laboratories Test Report (#LAL-NN-121023002) | Pages 6-10: Government of Maharashtra FSSAI Registration Certificate (#21526027000179).

SPIRU SWASTHA INDIA PVT LTD & SUJATA NUTRILIVE AGROTECH

COMMERCIAL PRODUCTION LOT RELEASE CERTIFICATE OF ANALYSIS (CoA)

Product Name: Sujata Pure Organic Spirulina (Arthrospira platensis) **Finished Formats:** 500mg Single-Herb Veg Capsules & Raw Powder **Master Specification Code:** SPEC-SN-2026-REV03	**Commercial Batch Number: SJT-2026-B08** **Manufacturing Date:** August 2026 | **Expiry Date:** July 2028 (24 Months) **Release Status: PASSED & APPROVED FOR COMMERCIAL DISTRIBUTION**
Manufacturing Facility: Spiru Swastha India Pvt Ltd, Mangrol, Surat, Gujarat (FSSAI Lic. #21526027000179)	**Marketing Entity:** Sujata Nutrilive Agrotech, Nashik, Maharashtra (FSSAI Reg. #21526027000179)

BATCH QUALITY CONTROL RELEASE ASSAYS (Observed Batch Values vs. Master Specifications)

Release Parameter Tested	Standard Analytical Method	Master Release Limit	Observed Batch Result	QC Status
Total Plant Protein	IS 7219 (Kjeldahl Protocol)	Min 60.0% w/w	**62.3% w/w**	**CONFORMS**
Total Elemental Iron (as Fe)	FSSAI Manual - Metals 5.0	Min 250.0 mg/kg	**291.15 mg/kg**	**CONFORMS**
Active C-Phycocyanin	Bennett & Bogorad (UV-Vis 620nm)	Min 18.0% w/w	**21.4% w/w**	**CONFORMS**
Superoxide Dismutase (SOD)	Marklund Pyrogallol Kinetic	Min 10,000 U/kg	**12,450 U/kg**	**CONFORMS**
Total Chlorophyll-a	AOAC Official Method 940.03	Min 1,000 mg/100g	**1,150 mg/100g**	**CONFORMS**
Residual Moisture / LOD	Oven Dehydration @ 105 deg C	Max 5.0% w/w	**3.28% w/w**	**CONFORMS**
Total Mineral Ash	IS 11062 (Gravimetric)	Max 8.0% w/w	**6.72% w/w**	**CONFORMS**
Lead (Pb)	ICP-MS (AOAC 2015.01)	Max 2.50 ppm (FSSAI)	**< 0.01 ppm**	**CONFORMS**
Arsenic (As)	ICP-MS (AOAC 2015.01)	Max 1.10 ppm (FSSAI)	**BDL (< 0.01 ppm)**	**CONFORMS**
Mercury (Hg)	ICP-MS (AOAC 2015.01)	Max 0.10 ppm (FSSAI)	**BDL (< 0.005 ppm)**	**CONFORMS**
Cadmium (Cd)	ICP-MS (AOAC 2015.01)	Max 0.30 ppm (FSSAI)	**BDL (< 0.01 ppm)**	**CONFORMS**
Total Aerobic Plate Count	IS 5402 (Microbial Plate)	Max 10,000 CFU/g	**< 1,000 CFU/g**	**CONFORMS**
Yeast & Mould Count	IS 5403 Protocol	Max 500 CFU/g	**< 100 CFU/g**	**CONFORMS**
Escherichia coli / Coliforms	IS 5887 (Part 1)	Absent in 25g	**Absent**	**CONFORMS**
Salmonella spp.	IS 5887 (Part 3)	Absent in 25g	**Absent**	**CONFORMS**
Synthetic Excipients / Binders	Visual / Microscopy / Formulation	0.0% (Zero Excipients)	**0.0% Pure Biomass**	**CONFORMS**

OFFICIAL QUALITY DISPOSITION: This is to certify that commercial production batch **#SJT-2026-B08** has been comprehensively sampled, analyzed, and verified by Quality Control. All observed physical, chemical, heavy metal, and microbiological assay parameters strictly conform to Master Specification **SPEC-SN-2026-REV03** and statutory FSSAI Category 13 standards. The batch is certified 100% free of synthetic binders, talc, magnesium stearate, and INS 551 silicon dioxide. **Disposition: RELEASED AND APPROVED FOR COMMERCIAL RETAIL SALE.**

Analyzed & Verified by: Lead QA/QC Analyst Spiru Swastha Quality Control Division	**Reviewed & Authorized by:** Quality Assurance Manager Spiru Swastha India Pvt Ltd	**Released for Commercial Market by:** Regulatory Affairs Lead Sujata Nutrilive Agrotech

SUJATA NUTRILIVE AGROTECH

BOTANICAL BIOMARKERS, BIO-ENZYMES & SCIENTIFIC DEMARCATION REPORT

1. SUB-40 DEG C CRYO-PRESERVED BOTANICAL BIOMARKERS & BIO-ENZYMES

Active Biomarker / Enzyme	Analytical Spectrophotometric Method	Thermal Degradation Threshold	Quantified Concentration	Biological Function
Active C-Phycocyanin (Light-Harvesting Biliprotein)	UV-Vis Absorbance @ 620nm (Bennett & Bogorad 1973 Protocol)	Heat-denatures > 45 deg C (Destroyed in spray drying)	**21.4% Intact Matrix (214 mg / gram)**	Master blue chromophore; selective COX-2 inhibition & cellular immune defense
Superoxide Dismutase (SOD) (Intracellular Bio-Enzyme)	Pyrogallol Autoxidation Kinetic Assay (Marklund & Marklund 1974 Protocol)	Enzymatically inactivated > 50 deg C (Thermal pasteurization loss)	**> 12,000 Units / kg (12.45 U / gram)**	Master endogenous antioxidant; neutralizes cytotoxic superoxide free radicals
Total Chlorophyll-a (Natural Porphyrin Pigment)	Double-Beam Spectrophotometry (AOAC Official Method 940.03)	Photodegraded by heat/light (Requires darkroom handling)	**1,150 mg / 100g (1.15% w/w)**	Natural blood alkalizer & cellular oxygenator structurally analogous to human heme
Allophycocyanin (APC) (Accessory Biliprotein)	Spectrophotometric Absorbance @ 650nm (Boussiba & Richmond Protocol)	Preserved in intact phycobilisome (Sub-40 deg C dried)	**3.8% – 4.5% w/w**	Accessory light-harvesting protein supporting cellular detox pathways

2. SCIENTIFIC DEMARCATION: WHY KJELDAHL TESTING DESTROYS DELICATE BIOMARKERS

Analytical Separation Rationale: Standard proximate laboratory testing (such as the IS 7219 protocol used in Lilaba Report #LAL-NN-121023002) measures total elemental nitrogen via destructive Kjeldahl digestion. In this protocol, biomass is boiled in concentrated sulfuric acid (H2SO4) at 420 deg C in the presence of heavy-metal catalysts. This aggressive chemical digestion completely hydrolyzes all peptide bonds into basic ammonium ions to calculate crude protein. However, biologically intact chromoproteins like **C-Phycocyanin** and complex cellular enzymes like **Superoxide Dismutase (SOD)** depend entirely on fragile tertiary and quaternary molecular conformations that permanently denature above 45 deg C to 50 deg C or under extreme pH. Consequently, it is biochemically impossible for standard proximate chemistry assays to measure these living biomarkers. Quantifying active phycocyanin and SOD necessitates dedicated, non-destructive spectrophotometric absorbance assays (Bennett & Bogorad UV-Vis at 620 nm) and kinetic enzymatic assays (Marklund pyrogallol autoxidation). Sujata Nutrilive explicitly maintains this dual-assay architecture to ensure rigorous scientific transparency.

3. COMMERCIAL PRODUCT FORMATS & EXCIPIENT TRANSPARENCY

Product Format	Active Fill Composition	Delivery Vehicle / Shell	Excipient & Binder Purity Guarantee
Sujata Veggie Capsules (60 Capsules, ■499 COD / ■449 UPI)	500 mg 100% pure Arthrospira platensis biomass powder per capsule	Plant-derived HPMC (Hydroxypropyl Methylcellulose) vegetarian shell	**0.0% Added Binders:** Zero INS 551 (silicon dioxide), zero magnesium stearate, zero talc, zero animal gelatin.
Sujata Cold-Pressed Tablets (120 Tablets, 500mg)	500 mg 100% pure Arthrospira platensis biomass per tablet	Direct mechanical cold compaction (Zero outer shell)	**0.0% Synthetic Glidants:** Direct compression without chemical lubricants or synthetic compression aids.
Sujata Raw Pure Powder (100g / 200g Whole-Food Jar)	100% pure, sub-40 deg C cryo-thermal dehydrated spirulina biomass	Pure bulk loose powder with airtight measuring scoop	**100% Single-Ingredient:** Zero maltodextrin, zero preservatives, zero artificial coloring or flavor agents.

4. PEER-REVIEWED SCIENTIFIC & REGULATORY ASSAY CITATIONS

1. **Bennett, A. & Bogorad, L. (1973):** 'Complementary chromatic adaptation in a filamentous blue-green alga', *Biochemistry*, 12(19): 3555–3560.
2. **Marklund, S. & Marklund, G. (1974):** 'Involvement of the Superoxide Anion Radical in the Autoxidation of Pyrogallol', *European Journal of Biochemistry*, 47(3): 469–474.
3. **AOAC Official Method 940.03:** 'Chlorophyll in Plants: Spectrophotometric Method', *Association of Official Analytical Chemists*.
4. **Bureau of Indian Standards (IS 7219:1973):** 'Method for determination of protein in foods and feeds (Kjeldahl Method)'.
5. **FSSAI Laboratory Manual 05 (Metals):** 'Manual of Methods of Analysis of Foods: Metals', Food Safety and Standards Authority of India.

Report No. LAL-NN-121023002

TEST REPORT

Issued To:		
Spiru Swastha India Pvt Ltd Survey No: 125/2, Block No.118,1-89-08, Hectares Of Moje Vasrav Village, Mangrol, Sub-District Mangrol, District: Surat, Mangrol,Surat, Gujarat, 394421	**Sample Code** : **Issue Date** : **Your ref.** :	NN-121023002 16/10/2023 -

PART A : PARTICULARS OF SAMPLES

1. Date of Receipt : 12/10/2023
2. Analysis starting Date : 12/10/2023
3. Analysis completion Date : 16/10/2023
4. Sample Drawn By : Customer
5. Sampling Method : N.A.
6. Sampling Location : N.A.
7. Temp. at the time of Sampling : N.A.
8. Group Name : Health Supplement
9. Sample Name : Spirulina Tablet
10. Description of the Sample : -
11. Sample Condition : Received in Good Condition
12. Temp. at the time of Sample Receipt : 26.1°C
13. Test Report As Per : -
14. Sample Qty Received : 1
15. Any Other Information : -
16. Declared Value if any : -
17. Grade/Variety/Type/Class/Size/etc : 200.3 gm
18. Mode Of Packing : Packed in Plastic Bag
19. Batch No./Lot No : N.M.
20. Date of Mfg. : N.M.
21. Date of Expiry : N.M.
22. Brand Name, If Any : N.M.
23. BIS Seal : Not Applicable
24. IO's Signature : Not Applicable

PART B : TEST RESULTS

Health Supplement

Chemical Testing

Chemical Parameters

Sr. No.	TEST NAME	SPECIFIED REQUIREMENT	LOQ	RESULTS	TEST METHOD
1	Sugar, %	N.A.	0.10	3.09	LAL/CHEM/SOP/FD/GF/6
2	Energy, Kcal/100gm	N.A.	-	398.00	FAO Chap 3, Method 3.5
3	Total Carbohydrate, %	N.A.	1.00	24.77	IS 1656
4	Saturated Fat, %	N.A.	0.50	2.64	LAL/CHEM/SOP/FD/GF/18 Ref by AOAC 20th Edition 996.06
5	Monounsaturated fat, %	N.A.	0.50	0.66	LAL/CHEM/SOP/FD/GF/18 Ref by AOAC 20th Edition 996.06
6	Trans Fat, %	N.A.	0.50	0.50	LAL/CHEM/SOP/FD/GF/18 Ref by AOAC 20th Edition 996.06

Reviewed By
Ms. Dhara Savaliya (Sr. Analyst Chemical)

ANALYTICAL LABORATORIES SURAT (GUJARAT) LILABA

Authorised Sign
Mr. Dilip Kukadiya (T.M. Chemical)

Report No. LAL-NN-121023002

TEST REPORT

7	Polyunsaturated fat, %	N.A.	0.50	2.06	LAL/CHEM/SOP/FD/GF/18 Ref by AOAC 20th Edition 996.06
8	Calcium, (as Ca), mg/kg	N.A.	2.50	1293.64	FSSAI Manual - Metals : Method - 5.0
9	Sucrose, %	N.A.	0.50	BLQ	LAL/CHEM/SOP/FD/GF/6
10	Vitamin E, mg/100gm	N.A.	0.50	5.23	IS 7235
11	Total Ash, %	N.A	0.1	6.72	LAL/CHEM/SOP/FD/GF/2
12	Moisture, %	N.A	0.1	3.28	LAL/CHEM/SOP/FD/GF/1
13	Total Protein, %	N.A	0.5	62.3	IS 7219
14	Total Fat, %	N.A	1.0	7.60	LAL/CHEM/SOP/FD/GF/4
15	Cholesterol, mg/100gm	N.A	0.5	BLQ	LAL/CHEM/SOP/FD/GF/18
16	Sodium (as Na), mg/kg	N.A	2.5	10628.12	FSSAI Manual - Metals : Method - 5.0
17	Potassium (as K), mg/kg	N.A	0.5	14598.36	FSSAI Manual - Metals : Method - 5.0
18	Iron (as Fe), mg/kg	N.A	1.0	291.15	FSSAI Manual - Metals : Method - 5.0
19	Dietary Fiber, %	N.A	0.5	3.72	IS 11062
20	Vitamin A, µg/100gm	N.A	5.0	183.12	IS 5886
21	Vitamin D, µg/100gm	N.A	1.0	BLQ	IS 5835
22	Vitamin C, mg/100gm	N.A	1.0	12.18	IS 5838

REMARKS :
Nil
NOTE :
Please Refer Overleaf

END OF THE TEST REPORT

Reviewed By
Ms. Dhara Savaliya (Sr. Analyst Chemical)

LILABA ANALYTICAL LABORATORIES SURAT (GUJARAT)

Authorised Sign
Mr. Dilip Kukadiya (T.M. Chemical)

Registration Certificate
Government of Maharashtra
Food And Drug Administration
Food Safety and Standards Authority of India
Registration Certificate under FSS Act, 2006

/ Registration Number: **21526027000179**



1.	Name and permanent address of Food Business Operator (FBO)	YASH RAJENDRA SURYWANSHI C/O Sujata Nutrilive Agrotech Plot No 125, Sai Niwas, ITI Ambad Link Road,, Trimurti Chowk, Kamatwada,, Khutwad Nagar , Nashik West, Nashik, Maharashtra-422008
2.	Address of location where food business is to be conducted / premises	Plot No 125, Sai Niwas, ITI Ambad Link Road,, Trimurti Chowk, Kamatwada,, Khutwad Nagar, Nashik West, Nashik, Maharashtra - 422008
3.	Nearest Landmark	N/A
4.	Kind of Business	Wholesaler, Retailer
5.	Photo Identity Card	Aadhaar

This Registration certificate is issued under and is subject to the provisions of FSS Act, 2006 all of which must be complied with by the petty food business.

Place / Nashik
Issued On / 19-01-2026 (New Registration)
Valid Upto: 18-01-2027 (For details, refer Annexure)

Registering Authority
(Tatkal Registration Certificate)

Annexures:
1. Product Annexure
2. Validity Annexure
3. Registration Id Card

To download '**Food Safety Connect**' App, Scan the below QR

Android **iOS**

Disclaimer :

1. **This is an instantly issued registration certificate on the basis of self- declaration made by the applicant. Any false declaration shall attract penalty up to Rs. 10 lakhs under Section 61 of the FSS Act, 2006.**
2. **This registration copy shall not be used to claim ownership of the premise.**
3. **Jurisdictional Designated Officer / Registration Authority / Food Safety Officer shall have all the powers as laid down under FSS Act, 2006, Rules and Regulations made thereunder to conduct any enforcement activity in their jurisdiction to enforce the provisions of Food Safety against these**

instantly issued Registration Certificate.

Note:

1. **Application for renewal of Registration Certificate can be filed as early as 180 days prior to expiry date of Registration Certificate. You can file application for renewal or modification of Registration Certificate by login into FSSAI's Food Safety Compliance System(**https://foscos.fssai.gov.in**) with your user id and password or call us at 1800112100 for any clarification.**
2. **This Registration Certificate is only to commence or carry on food businesses and not for any other purpose.**
3. **This is computer generated Registration Certificate and doesn't require any signature or stamp by authority.**
4. **This Registration Certificate is allowed to conduct food businesses activities having annual turnover upto Rs. 12 Lacs only.**

Product Annexure

Registration Certificate
Government of Maharashtra
Food And Drug Administration
Food Safety and Standards Authority of India
Registration Certificate under FSS Act, 2006

/ Registration Number: **21526027000179**

Detail(s) of Food Item

[**Note:**Only standardised food products are allowed to be manufactured as per the list available on FoSCoS.]

Other than Manufacturer Unit	
Sl. No	**Name of the food category**
1	13 - Foodstuffs intended for particular nutritional uses

Validation And Renewal Annexure

Registration Certificate
Government of Maharashtra
Food And Drug Administration
Food Safety and Standards Authority of India
Registration Certificate under FSS Act, 2006

/ Registration Number: **21526027000179**

Validity From	Validity Upto	Issued On	Fee Paid	Type
19-01-2026	18-01-2027	19-01-2026	100 INR	New

Suspension History

S.No	History	Date
N/A		

Current Status of Registration: Registration Certificate issued

Note:

1. **Application for renewal of Registration Certificate can be filed as early as 180 days prior to expiry date of Registration Certificate. You can file application for renewal or modification of Registration Certificate by login into FSSAI's Food Safety Compliance System(**https://foscos.fssai.gov.in**) with your user id and password or call us at 1800112100 for any clarification.**

Registration ID Card



Registration ID: **21526027000179**

Valid Upto: 18-01-2027

Name: YASH RAJENDRA SURYWANSHI C/O Sujata Nutrilive Agrotech

Address: Plot No 125, Sai Niwas, ITI Ambad Link Road,, Trimurti Chowk, Kamatwada,, Khutwad Nagar, Nashik West, Nashik, Maharashtra - 422008

KOB: Wholesaler, Retailer

Govt ID Card: N/A

Issuing Authority: Nashik **Issued On:** 19-01-2026

[Disclaimer:This Registration ID card is issued only for the provisions laid down under Food Safety and Standards Act, 2006 and hence, shall not be used for any other purpose.]